

Assignment – 2

1) RNN vs Transformer (Differences, Advantages & Limitations)

RNN (Recurrent Neural Network) vs Transformer (Difference)

RNN(Recurrent Neural Network)	TRANSFORMER
It will read One word at a time , it handling the sequential data	It will read all words together, it came in 2017
it have slow speed	It have fast speed
It have limited memory, Uses less battery, Runs on cheaper hardware	It have better memory
It is difficult for long sentence	It is easy for long sentence
It accuracy is less	It accuracy is more
It is used in older AI systems	It is used in modern AI system
It forgets old information	It remember old information better
it understand language but not better than transformer	It understand language better
In this current output depends on earlier inputs , it predicting the next words	It Looks at everything and focuses on what's important
EX: old translation system	EX: chat GPT, Gemini , Claude

ADVANTAGES

➤ RNN :

- Remembers Previous Information
- It is Good for Sequential Data
- Lower computational cost.
- Suitable for real-time streaming applications.
- Works well on small devices.
- RNN made AI work.

➤ TRANSFORMER

- Processes all words simultaneously.
- Trains much faster.

- Understands long-range relationships.
- Handles large datasets effectively.
- Forms the foundation of modern LLMs.
- Researchers introduced a paper called “Attention Is All You Need”
- Transformer made AI powerful

LIMITATIONS

➤ RNN

- Slow training.
- Difficulty remembering long information.
- Limited scalability.

➤ TRANSFORMER

- High memory usage.
- High training cost.
- Requires powerful hardware.
- Attention becomes expensive for very long inputs.

2)How Transformers Revolutionized AI

Revolutionized = Completely changed something for the better.

Transformers revolutionized AI means

Transformers completely changed AI and made it much more powerful.

Before transformer in before year 2017 , AI mostly used RNNs.

In this RNNs has a problem

Problem 1: Slow(One word at a time. This takes time.)

Problem 2: Forgetting(the information is too far away.)

Problem 3: Hard to Scale(RNNs became difficult to train when models grew larger.)

Then Transformers Arrived (2017) researches introduced a new idea .This is called Attention mechanism. The Transformer can look at all words together.

Why Was This a Big Breakthrough?

1. Much Faster Training
2. Better Understanding
3. Handles Long Documents(books , research papers much better than RNN)
4. Made LLMs Possible(Without Transformers , No ChatGPT, No Claude ,No Gemini)(Transformers are the technology that made modern LLMs possible.)
5. Better Translation
6. Better Coding AI(Transformers understand relationships between code words.)
7. Better Chatbots

Transformer is the **technology**.

LLM is the **model built using that technology**.

3)What are LLMs?

- LLM means **Large Language Model**
- It is an advanced in AI systems built on deep neural networks (DNNs)designed to process, understand and generate human-like text.
- By using massive datasets and billions of parameters, LLMs have transformed the way humans interact with technology.

Example

An LLM can:

- Answer questions
- Write stories
- Translate languages
- Generate code

Modern LLMs include ChatGPT (OpenAI), Google Gemini, Anthropic Claude, etc.

POPULAR LLMs

- GPT-4 and [GPT-4o](#) (OpenAI)
- [Gemini 1.5](#) (Google DeepMind): handles 1M+ tokens
- [Claude 3 \(Anthropic\)](#) (Safety-focused, strong at reasoning and summarization.)
- [LLaMA 3 \(Meta\)](#) (popular for startups)
- [Mistral 7B](#) / [Mixtral \(Mistral AI\)](#) (Efficient open-source alternatives for developers.)
- [BERT](#) and [RoBERTa](#) (Google/Facebook) (Strong embedding models for NLP tasks.)

APPLICATIONS

- Code Generation
- Debugging and Documentation
- Question Answering
- Language Translation and Correction
- Prompt-Based Versatility

ADVANTAGES

- It can perform new tasks using zero-shot and few-shot learning without retraining.
- It can understand and process the large amount of text data
- It can adapt easily and specific domains
- It can reduce human efforts on repetitive language-based tasks
- It Works on multiple domains like healthcare, education and business

LIMITATIONS

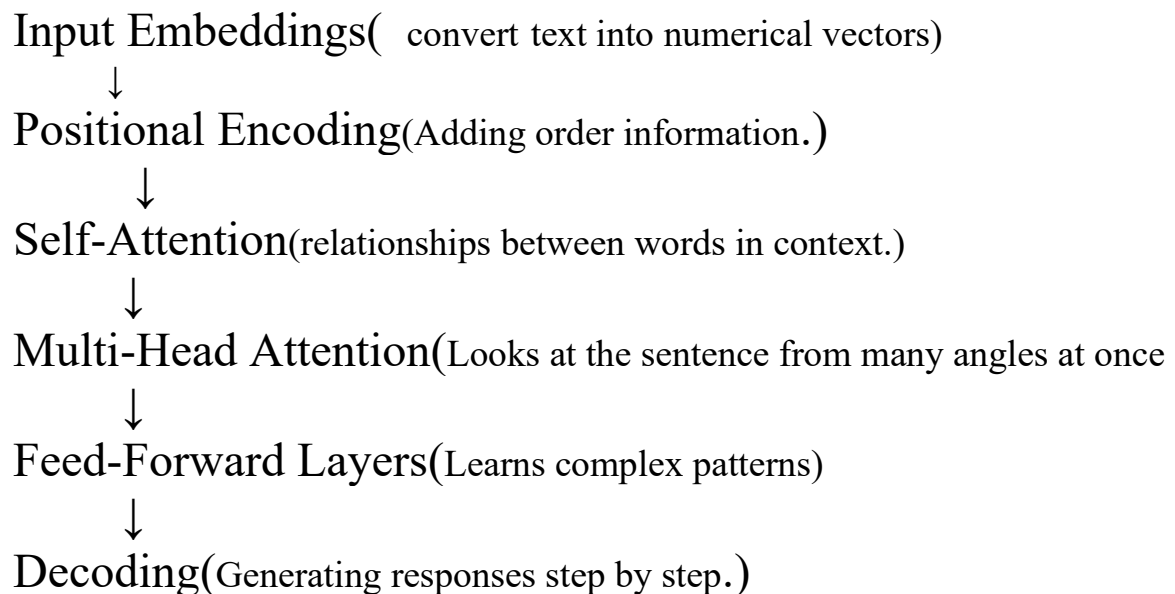
- it is expensive to train model.
- Training can take a long time.

- It depends on large amounts of high- quality and unbiased data
- It consume lot of electricity, it can affect environment

4)How LLMs Generate Text

- It based on transformer
 - asks a question.
 - AI reads the question.
 - AI predicts the next word.
 - Then predicts the next word again.
 - This continues until the answer is complete.

INPUT → EMBEDDING + POSITIONAL ENCODING → ENCODER BLOCK → DECODER BLOCK → OUTPUT



These components work together inside a Transformer to understand your input and generate meaningful text in an LLM.

Ex: the sky is “ “ it will predict next word

5. Parameters

Parameters are the AI's learned knowledge.

- They help the AI understand language.

More parameters usually mean:

- Better understanding
- Better answers

Ex: Think of parameters as tiny memories inside AI.

```
def add(x, y): # Here, x and y are parameters
    return x + y
```

6) Tokens

- A token is a small piece of text.

Ex: Sentence:

"I love AI"

Tokens:

- I
- love
- AI

Total = 3 tokens

AI reads tokens instead of full sentences.

7) Context Window

- A **Context Window** is the amount of text an AI can "remember" and use at one time while generating a response or conversation . Here AI is working memory
- Simple example we can remember it has white board , when board is long it can remember long information , when white board is small we want remove starting information to add new information

- It has two types **Small Context Window**(Remembers only a few pages.) , **Large Context Window**(Remembers many pages.)
- Bigger context = better memory

8)OPEN SOURCE VS CLOSED MODELS

OPEN SOURCE

- An open-source model that everyone can see.
- Anyone can use and modify
- More flexible.
- Usually free.

Examples:

- Meta's Llama models
- Mistral AI models
- Alibaba Group Qwen models
- DeepSeek

CLOSED SOURCE MODEL

- A closed-source model is like buying model.
- Only company controls it.
- More secure.
- Often paid.

Examples:

- OpenAI ChatGPT
- Google Gemini
- Anthropic Claude

9)LLM Benchmarks & Evaluation

When we create new AI model we want to check it is accurate, smart, better than older model that time we use this

- Benchmarks are tests used to measure AI performance. It is also used to compare different AI models

- Without benchmark we cannot know which model is better
- **Evaluation** means testing the AI model to see how well it performs. Like different tasks such as reasoning , mathematics, coding, and language.
- TYPES OF LLM BENCHMARKs
 1. Knowledge Test (it checks how much model knows)
 2. Reasoning Test (it checks thinking ability.)
 3. Math Test (Checks mathematical ability.)
 4. Coding Test (programming ability.)
 5. Reading Comprehension Test (whether the model understands passages.)
 6. Massive Multitask Language Understanding (MMLU)
- Therefore benchmarks help, but they do not tell the complete answer.

10) Prompt Engineering Fundamentals

Prompt means short pieces of text that are used to provide context and guidance to machine learning models.

- This help to generate relevant outputs which are as close to the expected output itself.
- Prompt engineering is similar - it's about crafting the right instructions, called prompts, to get the desired results from LLM

How prompt engineering works involves:

Crafting the Prompt: In this we design the prompt what LLM to do . It may be statement or question etc...

Understanding the LLM: Different prompts give different answers LLMs , the techniques involves LLM have minimal instructions zero shot prompting, few shot prompting

Refining the Prompt: it is trial – and- error process.

APPLICATION: It based more on text based modelling

- Language Translation(can translate the language)
- Question Answering Chatbots
- Text Generation
- In future the prompt engineering **AI and NLP, Integration with Other Technologies, Increased Automation and Efficiency**

11)AI Agents

AI Agents are smart AI systems that can perform tasks automatically.

- It operates with a degree of autonomy to complete assigned tasks effectively.
- An AI Agent can **do tasks**.
- It uses information and it will act
- It can work until goal is completed
- AI AGENT CYCLE
Goal (checking for dress under 1000rupees)

↓

Think

Think (it will ask itself)

↓

Plan

Plan (create a plan)

↓

Act

Act (perform action)

↓

Observe (check with goal)

↓

Improve (if does not match again it will check)

↓

Goal Completed (then it completed)

COMPONENTS OF AN AGENT

- Goal(with task want to complete)
- Memory (Stores information.)
- Reasoning (thinking process.)
- Tools (uses tools like calculator, email system, data base, search engines , **without tools agent only think , with tools it will act**)
- EX: Restaurant Agent, Study Agent, Travel Agent

They have Single-Agent (Only one AI handles everything.), Multi-Agent(Multiple AI agents work together.)

Why AI Agents Are Powerful? They can: Think , Plan ,Use tools, Make decisions, Remember information , Complete multi-step tasks

- Some tools used to build AI agents: [LangChain](#) ,[LangGraph](#) ,[CrewAI](#) [AutoGen](#) .These help developers create agent systems.
- AI Agent = Goal + Thinking + Planning + Action + Result

EX: Virtual Assistant Agent(siri), Study Assistant Agent, Travel Planning Agent, Email Agent, Healthcare Agent etc.....

12) RAG Fundamentals

RAG means Retrieval-Augmented Generation.(RAG = Search + LLM + Answer)

It is a technique that combines:

- Retrieval (searching for information)
- Generation (creating an answer)
- Instead of answering from training data , it will first searches for relevant information and then generates a response.

RAG is created because in LLMs model have some problems like **Limited Knowledge, Outdated Information, Hallucinations**(model may generate incorrect answer confidently), **Private Data this all** solved by the **RAG**

Basic RAG Workflow

Step 1: User asks a question.("What is Artificial Intelligence?")

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Step 2: Search for relevant information.

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Step 3: Retrieve useful documents.

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Step 4: Send documents to the LLM.

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Step 5: LLM generates the final answer.

Main Components of RAG: User Query(user ask question), Knowledge Base(collection of information), Retriever(finding relevant information), LLM(Reads the retrieved information and generates an answer.)

- **Retrieval** : finding relevant data
- **Generation:** Generation means creating a response using the retrieved information.
- **Embeddings:** Words are converted into numbers called embeddings.

Benefits of RAG

- More accurate answers
- Updated information
- Less hallucination
- Uses private data
- No retraining required

Limitations of RAG

- Retrieval may fail
- Poor documents give poor answers
- Requires additional infrastructure
- Slightly slower than direct LLM responses

EX: Chatbots, search systems, and enterprise AI.

13)Multimodal AI

- Multimodal AI can understand different types of data at a same time.
- Multimodal AI = Text + Images + Audio + Video
- "Multi" means many. "Modal" means type of data.

How Multimodal AI Works

1. Receives different types of input.
2. Understands each input.
3. Combines the information.
4. Generates a response.

Applications

- Healthcare
- Education
- Self-Driving Cars
- Virtual Assistants

Advantages

- Better understanding of information
- More accurate responses
- More natural human-computer interaction
- Supports multiple input types

Limitations

- Requires large amounts of data
- More complex than text-only AI
- Higher computing cost

EX: Gemini, claude, chatgpt, Vision-Language Models (VLMs), AI video understanding, Multimodal RAG

14)Model Fine-Tuning

Fine-tuning means giving extra training to an already trained AI model for a specific task.

Pre-trained Model + New Data = Fine-Tuned Model

Why Fine-Tuning?

- Improves accuracy
- Learns domain-specific knowledge
- Adapts to business needs
- Produces better responses

Example

General Model: Knows many topics.

Fine-Tuned Model:

Becomes an expert in: Healthcare, Finance, Coding ,Customer Support

Fine-Tuning Process

1. Pre-trained model
2. Collect task-specific data
3. Train the model
4. Get improved model

Advantages

- Better performance
- Customized responses
- Domain expertise
- Improved accuracy

Limitations

- Needs training data
- Requires computing resources
- Can be costly

Applications

- Medical Chatbots
- Coding Assistants
- Legal AI
- Banking AI
- Customer Support Bots

15) Why Enterprises Use Multiple Models

Enterprises use multiple AI models because one model cannot perform every task efficiently.

Enterprises use multiple AI models to improve accuracy, reduce cost, handle specialized tasks, and increase reliability.

Reasons

- Better Performance (Different models are good at different tasks.)
- Cost Savings (Small tasks use smaller, cheaper models.)
- Higher Accuracy (Best model can be chosen for each task.)
- Reliability (If one model fails, another model can be used.)

Specialized Tasks: Different models for:

- Text , Images ,Audio , Coding

Benefits

- Improved efficiency
- Reduced cost
- Better scalability
- Better user experience

EX: Chatbot → Language Model, Image Analysis → Vision Model, Speech Recognition → Audio Model

16) AI Safety, Privacy and Hallucinations

AI Safety:

AI Safety means ensuring AI systems behave correctly and do not cause harm.

Key Points

- Prevent harmful outputs
- Reduce bias and risks
- Ensure responsible AI use
- Improve reliability

EX: AI should not provide dangerous or misleading advice.

AI PRIVACY

AI Privacy means protecting user data and personal information.

Key Points

- Protect sensitive data
- Secure user information
- Prevent unauthorized access
- Follow privacy regulations

EX: A chatbot should not reveal a user's personal details.

AI Hallucinations

Hallucination occurs when AI generates incorrect or fake information but presents it as true.

Key Points

- Creates false information
- May sound confident
- Common in LLMs
- Reduced using RAG and verification

Ex : AI invents a book, fact, or website that does not exist.

- Recently focuses on Safer AI systems , Better privacy protection, Reduced hallucinations ,Human oversight, Responsible AI development

17) Comparative Analysis of Leading AI Models

Model	Company	Strength
ChatGPT	OpenAI	Reasoning, Coding, Agents
Gemini	Google	Multimodal, Google Integration
Claude	Anthropic	Long Documents, Safety
Llama	Meta	Open Source, Customization
Qwen	Alibaba	Strong Open Models, Coding
Mistral	Mistral AI	Efficient and Lightweight

ChatGPT (OpenAI) :-

- Agent capabilities
- Multimodal (text, image, voice)
- Best General Purpose

Gemini (Google)

- Strong multimodal AI
- Works with Google services
- Good for search and productivity
- Best Google Ecosystem

Claude (Anthropic)

- Excellent document analysis
- Focus on AI safety
- Handles large context windows
- Best Long Documents

Llama (Meta)

- Open-source model family
- Easy to customize
- Popular for enterprise deployment
- Best Open Source

Qwen (Alibaba)

- Strong coding performance
- Good multilingual support
- Popular open model
- Best Open-Source Coding

Mistral

- Fast and efficient
- Lower computing requirements

- Suitable for local deployment
- Best Lightweight Model

Leading AI models differ in strengths such as reasoning, coding, multimodal capabilities, safety, efficiency, and customization.

□ This are models Multimodal AI , AI Agents , Larger Context Windows ,Better Reasoning ,Open-Source Growth, Enterprise AI Adoption